

Life Cycles

Use the cue, then fade the cue. Study the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Living things grow and change in a repeating cycle. A **butterfly** has complete metamorphosis: egg → larva → pupa → adult. A **grasshopper** has incomplete metamorphosis: egg → nymph → adult. A **plant** grows: seed → seedling → mature plant → flower/seeds.

Round 1 — Retrieve It

1. What are the four stages of a butterfly's life cycle, in order?

2. What are the three stages of a grasshopper's life cycle, in order?

3. What grows first when a plant's life cycle begins?

4. Why is a life cycle called a “cycle” instead of a straight line?

Round 2 — Sketch It

Draw the butterfly life cycle as a circle. Show and label all four stages with arrows pointing from one stage to the next.



Life Cycles — Apply It

Keep the cue card covered. Answer first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Cycle

1. You find a caterpillar munching leaves. Which stage of the butterfly's life cycle is this, and what stage comes next?

2. A young grasshopper looks like a small adult with no wings yet. Which stage is it in, and what stage does it skip that a butterfly has?

3. A gardener plants a seed. List the stages the plant will go through before it can make new seeds.

Round 4 — Reason from Evidence

A student says all insects change the same way as they grow. Use the butterfly and the grasshopper to explain why this is not true.

Life Cycles — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain the difference between complete metamorphosis and incomplete metamorphosis. Name the stage that makes them different.

C**Connect**

Connect an animal life cycle to a plant life cycle. Name one way they are alike and one way they are different.

E**Extend**

Imagine a new insect. Draw or describe its life cycle stages in order, and tell whether it has complete or incomplete metamorphosis.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Life Cycles

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. Egg → larva (caterpillar) → pupa (chrysalis) → adult.
2. Egg → nymph → adult.
3. A seed sprouts first (then a seedling grows).
4. Because it repeats: adults make eggs or seeds that grow into new living things, which start the stages over again.

Sketch It — look for

A circle with all four butterfly stages labeled in order (egg, larva, pupa, adult) and arrows showing the cycle returning to the egg.

Apply It

1. It is the larva stage; the pupa (chrysalis) comes next.
2. It is the nymph stage; the grasshopper skips the pupa stage that a butterfly has.
3. Seed → seedling → mature plant → flower/seeds.

Reason from Evidence — look for

Not all insects change the same way. A butterfly has complete metamorphosis (four stages with a pupa), but a grasshopper has incomplete metamorphosis (three stages, no pupa, and nymphs look like small adults).

ACE — Extend look-for

Stages are listed in a sensible order and the student correctly labels the type: complete metamorphosis includes a pupa stage; incomplete does not.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color.